



MERIDIAN JUNIOR COLLEGE
JC2 Preliminary Examinations 2015
Higher 2

CANDIDATE
NAME

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CIVICS
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INDEX
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H2 BIOLOGY

9648/03

Applications Paper and Planning Question

21 September 2015

Paper 3

2 hours

Additional Materials: Answer papers

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

Write your name, civics group and index number on all the work you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use a soft pencil for any diagrams, graphs or rough workings.

Do not use staples, paper clips, highlighters, glue or correction fluid/tape.

At the end of the examination,

1. Fasten all your work securely together.
2. Hand in the following separately:
 - Structured questions 1 – 3
 - Planning question
 - Essay question

The number of marks is given in brackets [] at the end of each question or part question.

For examiner's Use	
1	/ 17
2	/ 10
3	/ 13
4 Planning	/ 12
5 Essay	/ 20
Total	/ 72

This paper consists of **10** printed pages.

[Turn over]

QUESTION 1

Golden Rice™ is a genetically modified form of rice that produces relatively large amounts of beta-carotene in the endosperm. Beta-carotene is metabolised in the human body to produce vitamin A.

- a) Explain why rice has been genetically modified to produce extra beta-carotene. [2]

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- b) The first types of Golden Rice™ produced only a very low mass of beta-carotene per gram of rice. Research continued to try to increase this.

Fig. 1.1 shows the metabolic pathway by which beta-carotene is synthesised in plants, and the enzymes that catalyse each step of the pathway.

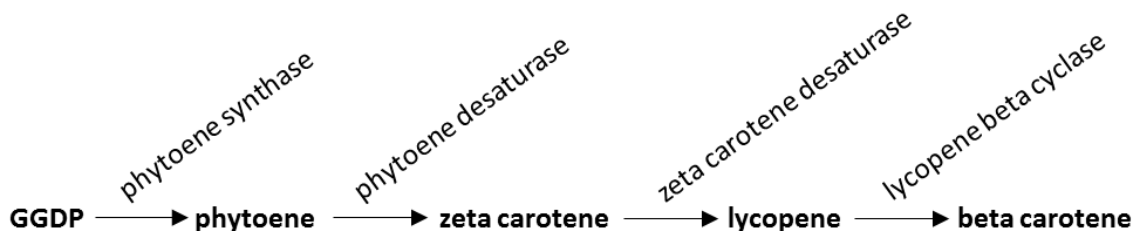


Fig. 1.1

The first types of Golden Rice™ contained a phytoene synthase gene, *psy*, from daffodils and a gene *crtl*, which produced the two desaturase enzymes, from the bacterium *Erwinia uredovora*.

Measurements of the quantities of intermediates in this metabolic pathway in rice endosperm showed that there was always a large amount of GGDP present, and that no phytoene accumulated in the tissues.

Explain how this suggests it was not the enzymes produced by the *crtl* gene that were limiting the production of beta-carotene. [2]

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c) Investigations were carried out to see if *psy* genes taken from species other than daffodils would enable rice endosperm to produce greater quantities of beta-carotene than the first types of Golden Rice™.

- *Psy* genes were isolated from the DNA of maize, tomatoes, peppers and daffodils. The genes were inserted into different plasmids.
- The promoter *Ubi1* from plants, and *crtl* genes from *E. uredoovora*, were also inserted into all of the plasmids.
- The four types of genetically modified plasmids were then separately inserted into different calluses of rice.
- The quantity of beta-carotene produced by these rice calluses was measured.

The results are shown in Table 1.1.

Source of <i>psy</i> gene	Total beta-carotene content of rice cells / arbitrary units
maize	14
pepper	4
tomato	6
daffodil	1

Table 1.1

i) Describe the features of calluses. [2]

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ii) Outline how calluses of rice were obtained. [2]

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iii) Explain why *Ubi1* promoter was inserted into the plasmids. [2]

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iv) Explain if results in Table 1.1 support the hypothesis that the *psy* gene, not the *crtl* gene, was limiting the production of beta-carotene in genetically modified rice. [2]

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d) After the production of the first batch of Golden Rice™ plants, plant tissue culture can continue to be used to generate more Golden Rice™ plants.

i) State one advantage of using plant tissue culture to propagate Golden Rice™ plants. [1]

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ii) Suggest why all plant tissue culture methods must be carried out under sterile conditions. [2]

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e) An agreement has been made between the commercial company that owns the production rights of golden rice and its developers. This allows the developers to give the rice to government-run breeding centres in rice-dependent countries. This agreement has addressed some concerns that initially arose from growing Golden Rice™.

State these concerns that initially arose from growing Golden Rice™. [2]

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[Total: 17]

QUESTION 2

Restriction fragment length polymorphism (RFLP) markers identified on the chromosomes are widely used to locate genes along the chromosomes.

In a species of plant, many RFLPs have been mapped within the genome. An RFLP analysis was used to map a herbicide-resistant gene on chromosome 2.

The F_1 offspring from crossing a homozygous herbicide-resistant (R) plant with a homozygous herbicide-sensitive (S) plant were self-fertilized to produce F_2 offspring. The F_2 offspring were analysed with regard to their herbicide sensitivity and two RFLP markers. Each RFLP marker has two variants. The result is shown in Fig. 2.1.

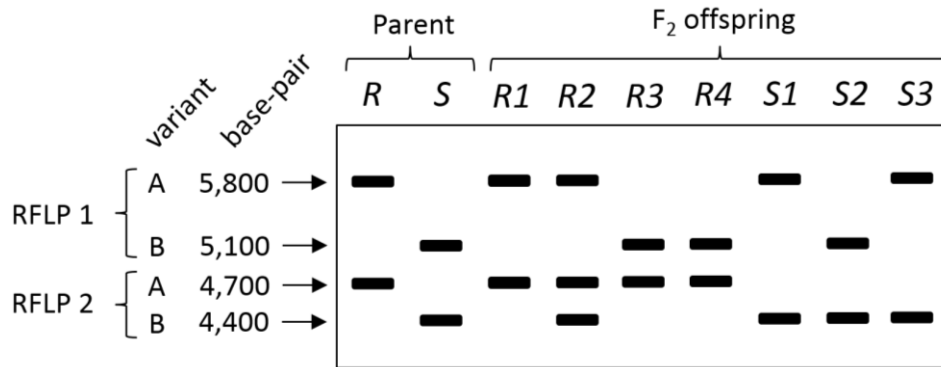


Fig. 2.1

- a) State and explain to which RFLP marker is the gene locus for herbicide-resistance likely to be tightly linked. [2]

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- b) The distance between two loci is determined by crossing a heterozygote with a homozygote. Fig. 2.2 shows the distance (in terms of map units) between the RFLP loci 1 and 2 on chromosome 2.

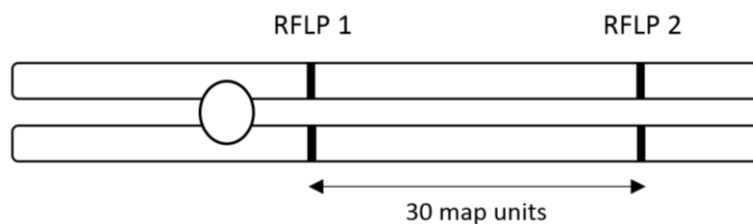


Fig. 2.2

State how the relative distance between the RFLP loci 1 and 2 is determined to be 30 map units. [1]

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In polymerase chain reaction, *Taq* DNA polymerase functions during the elongation step.

- c) Outline the role played by other components in a PCR mixture. [3]

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- d) Suggest how data from the Human Genome Project has facilitated the amplification of target DNA sequence using PCR. [1]

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- e) Scientists have found a new method of copying DNA that is faster than PCR. The new method, called helicase-dependent amplification (HDA), uses the enzyme helicase to separate the two strands of DNA. This means that DNA can be copied at a constant temperature of 37°C. In all other mechanical aspects, HDA works in exactly the same way as PCR.

- i) Explain why HDA will not work with *Taq* DNA polymerase. [2]

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- ii) Explain why HDA is faster than PCR in amplifying DNA. [1]

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[Total: 10]

QUESTION 3

Induced pluripotent stem (iPS) cells are adult cells that have been genetically reprogrammed to an embryonic stem cell-like state. Genetic reprogramming is carried out by using modified retroviruses to introduce genes for transcription factors into adult cells from a patient (Fig. 3.1).

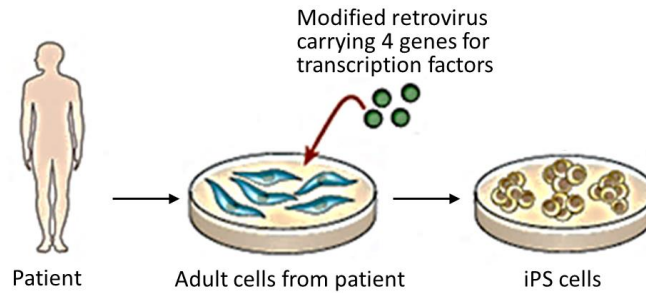


Fig. 3.1

- a) Explain the role of the transcription factors in genetic reprogramming of the adult cells. [2]

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- b) Apart from insertion of the four genes for transcription factors, describe two other ways in which the retroviral vector may be modified before they are used for genetic reprogramming of specific adult cells. [2]

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- c) Given that iPS cells are pluripotent like embryonic stem cells, they are promising tools for cell replacement therapy and drug development.

- i) Explain two advantages of using iPS cells for cell replacement therapy. [2]

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- ii) Suggest how iPS cells may be used for drug development to treat diseases such as cystic fibrosis. [2]

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Cystic fibrosis is caused by a mutation to the CFTR gene which encodes the Cystic Fibrosis Transmembrane Regulator (CFTR) protein. The CFTR protein regulates the transport of both chloride (Cl^-) and hydrogencarbonate ions (HCO_3^-) across the cell surface membrane. The secretory epithelial cells of the pancreatic duct and airway share the ability to generate HCO_3^- rich fluid.

A study is carried out to determine the effect of 10 different mutant CFTR alleles produced from a point mutation. The transport of HCO_3^- by epithelial cells expressing two copies of the normal CFTR allele was compared with that by epithelial cells expressing two copies of the same mutant CFTR allele. The results are shown in Table 3.1.

In the table, normal digestive functioning of the pancreas associated with a particular allele is indicated by a “+” and absence of normal functioning by a “–”.

CFTR allele		Percentage HCO_3^- transport in comparison with normal allele	Normal digestive functioning of pancreas
Normal		100	+
Mutant	1	6	–
	2	4	–
	3	0	–
	4	3	–
	5	1	–
	6	33	+
	7	41	+
	8	46	+
	9	37	+
	10	44	+

Table 3.1

- d) With reference to Table 3.1, describe the differences observed between different mutant CFTR alleles. [2]

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- e) Due to altered HCO_3^- secretion, tissues that express some mutant alleles secrete acidic fluids, while tissues that express normal CFTR alleles secrete alkaline fluids.

- i) State the mutant alleles that lead to these acidic secretions. [1]

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- ii) Briefly explain how the mutated alleles lead to acidic secretions. [2]

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[Total: 13]

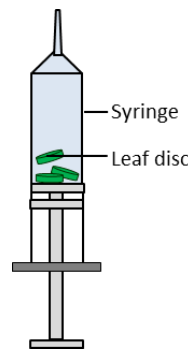
QUESTION 4 – Planning question

You are required to plan, but not carry out an investigation into the effect of carbon dioxide on the rate of photosynthesis.

The spongy mesophyll layer of leaves is normally infused with gases (O_2 and CO_2), and hence discs cut from leaves will float on water. To carry out this experiment effectively, it is important for the leaf discs to sink, which can occur when the gases are forced out from the spongy mesophyll layer using a vacuum. A vacuum can be created by using a finger to close the nozzle of the syringe that is filled with solution. At the same time, draw the plunger to remove any gas that is in the spongy mesophyll layer. Once done, the leaf disc will be able to sink.

In the presence of bicarbonate ions, an alternate source of carbon dioxide, photosynthesis can occur in a sunken leaf disc. As photosynthesis proceeds, oxygen accumulates in the air spaces of the spongy mesophyll, and the leaf disc will once again become buoyant and rise in a column of water. The rate of photosynthesis can therefore be measured. Though some oxygen produced may be used in aerobic respiration, this amount used is negligible and can be ignored.

You are given pieces of fresh leaves, which you need to use along with the set up shown below. Each piece of fresh leaf can only give 10 leaf discs.



Your planning must be based on the assumption that you have been provided with the following apparatus and materials which you must use:

- 8 pieces of fresh spinach leaves
- 0.2 gdm^{-3} Sodium hydrogen carbonate ($NaHCO_3$)
- distilled water
- core borer
- bench lamp which emit little heat
- 5cm^3 syringe
- timer, e.g. stopwatch or stop clock
- plasticine
- meter ruler
- Normal laboratory glassware, e.g. test-tubes, beakers, measuring cylinders, graduated pipettes, glass rods, etc.

Your plan should:

- have a clear and helpful structure such that the method you used is able to be repeated by anyone reading it,
- be illustrated by relevant diagrams, if necessary,
- identify the independent and dependent variables,
- describe the method with scientific reasoning used to decide the method so that the results areas accurate and reliable as possible,
- show how you will record your results and the proposed layout of results tables and graphs,
- use the correct technical and scientific terms,
- include reference to safety measures to minimise any risks associated with the proposed experiment.

[Total: 12]

Free-response question

For
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Use

Write your answer to this question on the separate answer paper provided.

Your answer:

- should be illustrated by large, clearly labeled diagrams, where appropriate;
- must be in continuous prose, where appropriate;
- must be set out in sections **(a)**, **(b)** etc., as indicated in the question

QUESTION 5

- a)** Restriction fragment length polymorphism (RFLP) analysis facilitates the process of DNA fingerprinting. DNA fingerprints can be used for paternity testing and elucidating the murderer in a crime scene.

Outline how DNA fingerprinting is carried out **and** explain its theoretical basis. [12]

- b)** Explain why there has been limited success in using gene therapy to treat genetic diseases. [8]

[Total: 20]

• END OF PAPER 3 •